Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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November 9, 2026

1. **Let $A_1, A_2, \ldots, A_n$ be invertible square matrices of the same order. Express $(A_1 A_2 \cdots A_n)^{-1}$ in terms of $A_1^{-1}, A_2^{-1}, \ldots, A_n^{-1}$.**

   Using $(AB)^{-1} = B^{-1}A^{-1}$ successively,

$$\begin{aligned}(A_1 A_2 \cdots A_n)^{-1} &= A_n^{-1}(A_1 A_2 \cdots A_{n-1})^{-1} \\ &= A_n^{-1} A_{n-1}^{-1}(A_1 A_2 \cdots A_{n-2})^{-1} \\ &= \cdots \\ &= \boxed{A_n^{-1} A_{n-1}^{-1} \cdots A_2^{-1} A_1^{-1}}.\end{aligned}$$